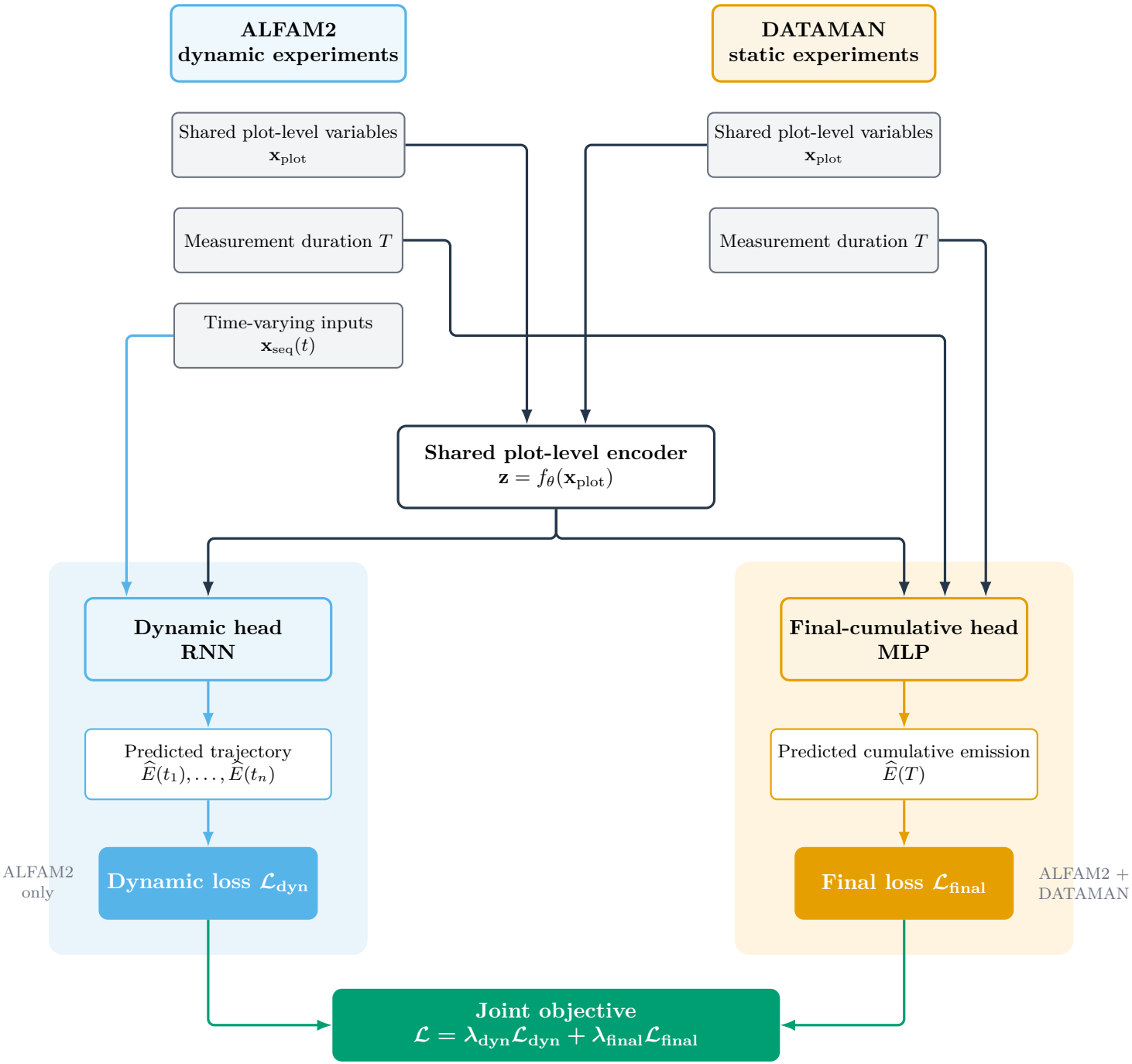
ALFAM2
dynamic experiments
DATAMAN
static experiments
Shared plot-level variables
$\mathbf{x}_{\text{plot}}$
Shared plot-level variables
$\mathbf{x}_{\text{plot}}$
Measurement duration T
Measurement duration T
Time-varying inputs
$\mathbf{x}_{\text{seq}}(t)$
Shared plot-level encoder
$\mathbf{z} = f_\theta(\mathbf{x}_{\text{plot}})$
Dynamic head
RNN
Final-cumulative head
MLP
Predicted trajectory
$\widehat{E}(t_1), \ldots, \widehat{E}(t_n)$
Predicted cumulative emission
$\widehat{E}(T)$
ALFAM2
only
Dynamic loss $\mathcal{L}_{\text{dyn}}$
Final loss $\mathcal{L}_{\text{final}}$
ALFAM2 +
DATAMAN
Joint objective
$\mathcal{L} = \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{final}}\mathcal{L}_{\text{final}}$